Huawei HCCDA-AI

5-Phase • 100 MCQ Exam Preparation

SECOND EDITION

100 Completely New Questions

Question → 4 Options → Correct Answer → Explanation of Every Option

Important: This is a fresh question bank. The questions are newly written for the 2nd Edition and are not copied from the previous 100-MCQ edition.

Basis: The uploaded Huawei HCCDA-AI course plan, including its AI foundations, ML/data work, ModelArts, Huawei Cloud EI APIs, OCR/Vision, HiLens, deep learning, testing, deployment, MLOps/CI-CD, and certification preparation topics.

Phase 1 — Artificial Intelligence (AI)

20 fresh questions in this phase.

Question 1: Which statement best distinguishes Artificial Intelligence from ordinary rule-based automation?

A. AI systems can perform tasks associated with intelligent behavior using learned or designed methods

B. AI is only a method for formatting documents

C. AI is limited to network configuration

D. AI refers only to computer hardware

Correct Answer: A. AI systems can perform tasks associated with intelligent behavior using learned or designed methods

Why this is correct: AI systems can perform tasks associated with intelligent behavior using learned or designed methods directly addresses the concept tested in the question.

Why B is incorrect: AI is only a method for formatting documents does not correctly answer the question.

Why C is incorrect: AI is limited to network configuration does not correctly answer the question.

Why D is incorrect: AI refers only to computer hardware does not correctly answer the question.

Question 2: Which category of AI refers to intelligence intended to perform a broad range of tasks comparable to human intelligence?

A. General AI

B. Narrow AI

C. Single-purpose automation

D. Database AI

Correct Answer: A. General AI

Why this is correct: General AI directly addresses the concept tested in the question.

Why B is incorrect: Narrow AI does not correctly answer the question.

Why C is incorrect: Single-purpose automation does not correctly answer the question.

Why D is incorrect: Database AI does not correctly answer the question.

Question 3: What is symbolic AI primarily associated with?

A. Explicit knowledge, rules, and logical reasoning

B. Only image pixels

C. Only cloud storage

D. Only hardware sensors

Correct Answer: A. Explicit knowledge, rules, and logical reasoning

Why this is correct: Explicit knowledge, rules, and logical reasoning directly addresses the concept tested in the question.

Why B is incorrect: Only image pixels does not correctly answer the question.

Why C is incorrect: Only cloud storage does not correctly answer the question.

Why D is incorrect: Only hardware sensors does not correctly answer the question.

Question 4: What is a key difference between symbolic AI and machine learning?

A. Symbolic AI commonly relies on explicit rules, while ML learns patterns from data

B. Symbolic AI always requires GPUs and ML never does

C. ML cannot use data

D. They are exactly the same approach

Correct Answer: A. Symbolic AI commonly relies on explicit rules, while ML learns patterns from data

Why this is correct: Symbolic AI commonly relies on explicit rules, while ML learns patterns from data directly addresses the concept tested in the question.

Why B is incorrect: Symbolic AI always requires GPUs and ML never does does not correctly answer the question.

Why C is incorrect: ML cannot use data does not correctly answer the question.

Why D is incorrect: They are exactly the same approach does not correctly answer the question.

Question 5: Why is AI industry adoption affected by cost?

A. AI solutions may require computing, data, infrastructure, and operational resources

B. Cost has no relationship to AI adoption

C. Cost changes image resolution only

D. Cost determines Python syntax

Correct Answer: A. AI solutions may require computing, data, infrastructure, and operational resources

Why this is correct: AI solutions may require computing, data, infrastructure, and operational resources directly addresses the concept tested in the question.

Why B is incorrect: Cost has no relationship to AI adoption does not correctly answer the question.

Why C is incorrect: Cost changes image resolution only does not correctly answer the question.

Why D is incorrect: Cost determines Python syntax does not correctly answer the question.

Question 6: Why can regulation affect AI adoption?

A. Organizations may need to meet legal, privacy, safety, or governance requirements

B. Regulation changes monitor brightness

C. Regulation replaces model training

D. Regulation determines the number of Python loops

Correct Answer: A. Organizations may need to meet legal, privacy, safety, or governance requirements

Why this is correct: Organizations may need to meet legal, privacy, safety, or governance requirements directly addresses the concept tested in the question.

Why B is incorrect: Regulation changes monitor brightness does not correctly answer the question.

Why C is incorrect: Regulation replaces model training does not correctly answer the question.

Why D is incorrect: Regulation determines the number of Python loops does not correctly answer the question.

Question 7: Which trend is associated with modern AI systems according to the course?

A. Large Language Models (LLMs)

B. Floppy disk storage

C. CRT displays

D. Manual typewriters

Correct Answer: A. Large Language Models (LLMs)

Why this is correct: Large Language Models (LLMs) directly addresses the concept tested in the question.

Why B is incorrect: Floppy disk storage does not correctly answer the question.

Why C is incorrect: CRT displays does not correctly answer the question.

Why D is incorrect: Manual typewriters does not correctly answer the question.

Question 8: What is the purpose of mapping a business use case to an AI capability?

A. To select an appropriate AI service, API, or model for the required task

B. To avoid defining inputs and outputs

C. To remove all testing requirements

D. To select a monitor

Correct Answer: A. To select an appropriate AI service, API, or model for the required task

Why this is correct: To select an appropriate AI service, API, or model for the required task directly addresses the concept tested in the question.

Why B is incorrect: To avoid defining inputs and outputs does not correctly answer the question.

Why C is incorrect: To remove all testing requirements does not correctly answer the question.

Why D is incorrect: To select a monitor does not correctly answer the question.

Question 9: Which item belongs in an AI application's scope definition?

A. The boundaries of what the system will and will not do

B. The color of the developer's laptop

C. The brand of keyboard

D. The office furniture layout

Correct Answer: A. The boundaries of what the system will and will not do

Why this is correct: The boundaries of what the system will and will not do directly addresses the concept tested in the question.

Why B is incorrect: The color of the developer's laptop does not correctly answer the question.

Why C is incorrect: The brand of keyboard does not correctly answer the question.

Why D is incorrect: The office furniture layout does not correctly answer the question.

Question 10: Why should user goals be identified during requirement analysis?

A. They help define what the AI system is expected to accomplish

B. They determine the monitor size

C. They replace technical requirements

D. They eliminate the need for stakeholders

Correct Answer: A. They help define what the AI system is expected to accomplish

Why this is correct: They help define what the AI system is expected to accomplish directly addresses the concept tested in the question.

Why B is incorrect: They determine the monitor size does not correctly answer the question.

Why C is incorrect: They replace technical requirements does not correctly answer the question.

Why D is incorrect: They eliminate the need for stakeholders does not correctly answer the question.

Question 11: What is an AI architecture sketch intended to communicate?

A. The major components and how they interact in the proposed system

B. Only the application's logo

C. Only the source-code comments

D. Only the final exam score

Correct Answer: A. The major components and how they interact in the proposed system

Why this is correct: The major components and how they interact in the proposed system directly addresses the concept tested in the question.

Why B is incorrect: Only the application's logo does not correctly answer the question.

Why C is incorrect: Only the source-code comments does not correctly answer the question.

Why D is incorrect: Only the final exam score does not correctly answer the question.

Question 12: Which AI domain is particularly relevant to extracting information from images of documents?

A. Computer Vision and OCR

B. Only networking

C. Only database indexing

D. Only operating systems

Correct Answer: A. Computer Vision and OCR

Why this is correct: Computer Vision and OCR directly addresses the concept tested in the question.

Why B is incorrect: Only networking does not correctly answer the question.

Why C is incorrect: Only database indexing does not correctly answer the question.

Why D is incorrect: Only operating systems does not correctly answer the question.

Question 13: Which AI domain is most directly associated with understanding and generating human language?

A. Natural Language Processing

B. Computer hardware design

C. Disk management

D. Network cabling

Correct Answer: A. Natural Language Processing

Why this is correct: Natural Language Processing directly addresses the concept tested in the question.

Why B is incorrect: Computer hardware design does not correctly answer the question.

Why C is incorrect: Disk management does not correctly answer the question.

Why D is incorrect: Network cabling does not correctly answer the question.

Question 14: Which AI domain is associated with interpreting spoken input?

A. Speech AI

B. Spreadsheet AI

C. Printer AI

D. Storage AI

Correct Answer: A. Speech AI

Why this is correct: Speech AI directly addresses the concept tested in the question.

Why B is incorrect: Spreadsheet AI does not correctly answer the question.

Why C is incorrect: Printer AI does not correctly answer the question.

Why D is incorrect: Storage AI does not correctly answer the question.

Question 15: What is a practical reason for conducting a pre-test before certification preparation?

A. It helps identify existing strengths and knowledge gaps

B. It permanently replaces the final exam

C. It removes the need for study

D. It changes the course duration

Correct Answer: A. It helps identify existing strengths and knowledge gaps

Why this is correct: It helps identify existing strengths and knowledge gaps directly addresses the concept tested in the question.

Why B is incorrect: It permanently replaces the final exam does not correctly answer the question.

Why C is incorrect: It removes the need for study does not correctly answer the question.

Why D is incorrect: It changes the course duration does not correctly answer the question.

Question 16: Why are sample questions useful during certification preparation?

A. They help learners practice the expected question style and reasoning process

B. They guarantee the exact final questions

C. They replace all practical work

D. They eliminate the need to understand concepts

Correct Answer: A. They help learners practice the expected question style and reasoning process

Why this is correct: They help learners practice the expected question style and reasoning process directly addresses the concept tested in the question.

Why B is incorrect: They guarantee the exact final questions does not correctly answer the question.

Why C is incorrect: They replace all practical work does not correctly answer the question.

Why D is incorrect: They eliminate the need to understand concepts does not correctly answer the question.

Question 17: What should a good AI problem statement identify?

A. The problem to be solved and the value expected from the solution

B. Only the programming language

C. Only the cloud provider logo

D. Only the developer's name

Correct Answer: A. The problem to be solved and the value expected from the solution

Why this is correct: The problem to be solved and the value expected from the solution directly addresses the concept tested in the question.

Why B is incorrect: Only the programming language does not correctly answer the question.

Why C is incorrect: Only the cloud provider logo does not correctly answer the question.

Why D is incorrect: Only the developer's name does not correctly answer the question.

Question 18: Why is a hybrid architecture sometimes considered for an AI application?

A. It can combine local/edge processing with cloud capabilities

B. It always removes the need for networking

C. It guarantees zero latency

D. It prevents cloud services from being used

Correct Answer: A. It can combine local/edge processing with cloud capabilities

Why this is correct: It can combine local/edge processing with cloud capabilities directly addresses the concept tested in the question.

Why B is incorrect: It always removes the need for networking does not correctly answer the question.

Why C is incorrect: It guarantees zero latency does not correctly answer the question.

Why D is incorrect: It prevents cloud services from being used does not correctly answer the question.

Question 19: What is one purpose of a requirement checklist?

A. To verify that important functional and non-functional requirements have been considered

B. To replace system testing

C. To delete incomplete requirements automatically

D. To choose a laptop model

Correct Answer: A. To verify that important functional and non-functional requirements have been considered

Why this is correct: To verify that important functional and non-functional requirements have been considered directly addresses the concept tested in the question.

Why B is incorrect: To replace system testing does not correctly answer the question.

Why C is incorrect: To delete incomplete requirements automatically does not correctly answer the question.

Why D is incorrect: To choose a laptop model does not correctly answer the question.

Question 20: Why should AI project requirements be reviewed by stakeholders?

A. Stakeholder review can reveal missing, unclear, or incorrect expectations

B. Stakeholders are only needed after deployment

C. Reviewing requirements changes CPU speed

D. Stakeholders replace the AI model

Correct Answer: A. Stakeholder review can reveal missing, unclear, or incorrect expectations

Why this is correct: Stakeholder review can reveal missing, unclear, or incorrect expectations directly addresses the concept tested in the question.

Why B is incorrect: Stakeholders are only needed after deployment does not correctly answer the question.

Why C is incorrect: Reviewing requirements changes CPU speed does not correctly answer the question.

Why D is incorrect: Stakeholders replace the AI model does not correctly answer the question.

Phase 2 — Machine Learning (ML)

20 fresh questions in this phase.

Question 1: What is supervised learning?

- **A.** Learning from examples that include target labels or expected outputs
- **B.** Learning without any data
- **C.** Only storing files in OBS
- **D.** Only monitoring a server

Correct Answer: A. Learning from examples that include target labels or expected outputs

Why this is correct: Learning from examples that include target labels or expected outputs directly addresses the concept tested in the question.

Why B is incorrect: Learning without any data does not correctly answer the question.

Why C is incorrect: Only storing files in OBS does not correctly answer the question.

Why D is incorrect: Only monitoring a server does not correctly answer the question.

Question 2: What is classification in machine learning?

- **A.** Predicting a categorical class or label
- **B.** Predicting monitor resolution
- **C.** Compressing a dataset
- **D.** Creating a cloud bucket

Correct Answer: A. Predicting a categorical class or label

Why this is correct: Predicting a categorical class or label directly addresses the concept tested in the question.

Why B is incorrect: Predicting monitor resolution does not correctly answer the question.

Why C is incorrect: Compressing a dataset does not correctly answer the question.

Why D is incorrect: Creating a cloud bucket does not correctly answer the question.

Question 3: What is regression used for?

- **A.** Predicting a continuous numerical value
- **B.** Assigning only image colors
- **C.** Creating API credentials
- **D.** Managing cloud storage

Correct Answer: A. Predicting a continuous numerical value

Why this is correct: Predicting a continuous numerical value directly addresses the concept tested in the question.

Why B is incorrect: Assigning only image colors does not correctly answer the question.

Why C is incorrect: Creating API credentials does not correctly answer the question.

Why D is incorrect: Managing cloud storage does not correctly answer the question.

Question 4: Why is feature preprocessing important?

A. It can transform raw data into a form more suitable for model training

B. It automatically guarantees perfect predictions

C. It replaces evaluation

D. It removes the need for a dataset

Correct Answer: A. It can transform raw data into a form more suitable for model training

Why this is correct: It can transform raw data into a form more suitable for model training directly addresses the concept tested in the question.

Why B is incorrect: It automatically guarantees perfect predictions does not correctly answer the question.

Why C is incorrect: It replaces evaluation does not correctly answer the question.

Why D is incorrect: It removes the need for a dataset does not correctly answer the question.

Question 5: What is the purpose of a validation set?

A. To help assess and tune a model during development

B. To store cloud passwords

C. To replace the training set

D. To record hardware inventory

Correct Answer: A. To help assess and tune a model during development

Why this is correct: To help assess and tune a model during development directly addresses the concept tested in the question.

Why B is incorrect: To store cloud passwords does not correctly answer the question.

Why C is incorrect: To replace the training set does not correctly answer the question.

Why D is incorrect: To record hardware inventory does not correctly answer the question.

Question 6: What is the purpose of a test set?

A. To provide a final assessment on data not used for model development

B. To update model weights continuously

C. To create training labels

D. To configure OBS

Correct Answer: A. To provide a final assessment on data not used for model development

Why this is correct: To provide a final assessment on data not used for model development directly addresses the concept tested in the question.

Why B is incorrect: To update model weights continuously does not correctly answer the question.

Why C is incorrect: To create training labels does not correctly answer the question.

Why D is incorrect: To configure OBS does not correctly answer the question.

Question 7: What does generalization mean in ML?

A. Performing well on previously unseen data

B. Memorizing every training example

C. Increasing dataset size automatically

D. Running only on a local laptop

Correct Answer: A. Performing well on previously unseen data

Why this is correct: Performing well on previously unseen data directly addresses the concept tested in the question.

Why B is incorrect: Memorizing every training example does not correctly answer the question.

Why C is incorrect: Increasing dataset size automatically does not correctly answer the question.

Why D is incorrect: Running only on a local laptop does not correctly answer the question.

Question 8: What is overfitting?

A. When a model fits training data too closely and performs poorly on unseen data

B. When a model has no training data

C. When an API returns JSON

D. When a dataset is stored in OBS

Correct Answer: A. When a model fits training data too closely and performs poorly on unseen data

Why this is correct: When a model fits training data too closely and performs poorly on unseen data directly addresses the concept tested in the question.

Why B is incorrect: When a model has no training data does not correctly answer the question.

Why C is incorrect: When an API returns JSON does not correctly answer the question.

Why D is incorrect: When a dataset is stored in OBS does not correctly answer the question.

Question 9: What is underfitting?

A. When a model is too simple to capture important patterns in the data

B. When a model memorizes every example

C. When an image is too large

D. When a cloud service is unavailable

Correct Answer: A. When a model is too simple to capture important patterns in the data

Why this is correct: When a model is too simple to capture important patterns in the data directly addresses the concept tested in the question.

Why B is incorrect: When a model memorizes every example does not correctly answer the question.

Why C is incorrect: When an image is too large does not correctly answer the question.

Why D is incorrect: When a cloud service is unavailable does not correctly answer the question.

Question 10: Why is class distribution important during model evaluation?

A. Uneven class distributions can make some metrics misleading and affect learning

B. Class distribution never affects ML

C. It only affects monitor size

D. It changes Python syntax

Correct Answer: A. Uneven class distributions can make some metrics misleading and affect learning

Why this is correct: Uneven class distributions can make some metrics misleading and affect learning directly addresses the concept tested in the question.

Why B is incorrect: Class distribution never affects ML does not correctly answer the question.

Why C is incorrect: It only affects monitor size does not correctly answer the question.

Why D is incorrect: It changes Python syntax does not correctly answer the question.

Question 11: What does precision measure?

A. The proportion of predicted positive cases that are actually positive

B. The proportion of all actual positives found

C. The number of training epochs

D. The number of model layers

Correct Answer: A. The proportion of predicted positive cases that are actually positive

Why this is correct: The proportion of predicted positive cases that are actually positive directly addresses the concept tested in the question.

Why B is incorrect: The proportion of all actual positives found does not correctly answer the question.

Why C is incorrect: The number of training epochs does not correctly answer the question.

Why D is incorrect: The number of model layers does not correctly answer the question.

Question 12: What does recall measure?

A. The proportion of actual positive cases that are successfully identified

B. The proportion of predicted positives that are correct

C. The size of the training dataset

D. The model's RAM usage

Correct Answer: A. The proportion of actual positive cases that are successfully identified

Why this is correct: The proportion of actual positive cases that are successfully identified directly addresses the concept tested in the question.

Why B is incorrect: The proportion of predicted positives that are correct does not correctly answer the question.

Why C is incorrect: The size of the training dataset does not correctly answer the question.

Why D is incorrect: The model's RAM usage does not correctly answer the question.

Question 13: Why might accuracy be insufficient as the only metric?

A. It can hide poor performance on minority classes or specific error types

B. Accuracy is never useful

C. Accuracy only measures latency

D. Accuracy measures cloud storage

Correct Answer: A. It can hide poor performance on minority classes or specific error types

Why this is correct: It can hide poor performance on minority classes or specific error types directly addresses the concept tested in the question.

Why B is incorrect: Accuracy is never useful does not correctly answer the question.

Why C is incorrect: Accuracy only measures latency does not correctly answer the question.

Why D is incorrect: Accuracy measures cloud storage does not correctly answer the question.

Question 14: What is a hyperparameter?

A. A configuration value chosen before or around model training, such as a learning rate

B. A label inside the test set

C. A cloud account password

D. A camera pixel

Correct Answer: A. A configuration value chosen before or around model training, such as a learning rate

Why this is correct: A configuration value chosen before or around model training, such as a learning rate directly addresses the concept tested in the question.

Why B is incorrect: A label inside the test set does not correctly answer the question.

Why C is incorrect: A cloud account password does not correctly answer the question.

Why D is incorrect: A camera pixel does not correctly answer the question.

Question 15: What is model tuning?

A. Adjusting model or training settings to improve validation performance

B. Renaming the project folder

C. Changing monitor brightness

D. Deleting evaluation data

Correct Answer: A. Adjusting model or training settings to improve validation performance

Why this is correct: Adjusting model or training settings to improve validation performance directly addresses the concept tested in the question.

Why B is incorrect: Renaming the project folder does not correctly answer the question.

Why C is incorrect: Changing monitor brightness does not correctly answer the question.

Why D is incorrect: Deleting evaluation data does not correctly answer the question.

Question 16: Why should prediction logs be collected in production?

A. They support performance analysis, troubleshooting, and monitoring

B. They automatically retrain every model

C. They replace source code

D. They remove the need for metrics

Correct Answer: A. They support performance analysis, troubleshooting, and monitoring

Why this is correct: They support performance analysis, troubleshooting, and monitoring directly addresses the concept tested in the question.

Why B is incorrect: They automatically retrain every model does not correctly answer the question.

Why C is incorrect: They replace source code does not correctly answer the question.

Why D is incorrect: They remove the need for metrics does not correctly answer the question.

Question 17: What is root-cause analysis in ML testing?

A. Investigating why a model or pipeline produced an observed failure

B. Counting Python files

C. Measuring monitor size

D. Renaming a dataset

Correct Answer: A. Investigating why a model or pipeline produced an observed failure

Why this is correct: Investigating why a model or pipeline produced an observed failure directly addresses the concept tested in the question.

Why B is incorrect: Counting Python files does not correctly answer the question.

Why C is incorrect: Measuring monitor size does not correctly answer the question.

Why D is incorrect: Renaming a dataset does not correctly answer the question.

Question 18: Why compare predictions against ground truth?

A. To determine how accurately the model performs on known expected outcomes

B. To increase CPU speed

C. To create cloud credentials

D. To change image formats

Correct Answer: A. To determine how accurately the model performs on known expected outcomes

Why this is correct: To determine how accurately the model performs on known expected outcomes directly addresses the concept tested in the question.

Why B is incorrect: To increase CPU speed does not correctly answer the question.

Why C is incorrect: To create cloud credentials does not correctly answer the question.

Why D is incorrect: To change image formats does not correctly answer the question.

Question 19: What is a retraining trigger?

A. A defined condition that causes a model training process to run again

B. A new keyboard shortcut

C. A document heading

D. A monitor refresh rate

Correct Answer: A. A defined condition that causes a model training process to run again

Why this is correct: A defined condition that causes a model training process to run again directly addresses the concept tested in the question.

Why B is incorrect: A new keyboard shortcut does not correctly answer the question.

Why C is incorrect: A document heading does not correctly answer the question.

Why D is incorrect: A monitor refresh rate does not correctly answer the question.

Question 20: Why should ML experiments be reproducible?

A. Reproducibility makes it easier to verify results and understand which changes affected performance

B. It prevents model evaluation

C. It removes the need for datasets

D. It guarantees a single correct model

Correct Answer: A. Reproducibility makes it easier to verify results and understand which changes affected performance

Why this is correct: Reproducibility makes it easier to verify results and understand which changes affected performance directly addresses the concept tested in the question.

Why B is incorrect: It prevents model evaluation does not correctly answer the question.

Why C is incorrect: It removes the need for datasets does not correctly answer the question.

Why D is incorrect: It guarantees a single correct model does not correctly answer the question.

Phase 3 — Deep Learning (DL)

20 fresh questions in this phase.

Question 1: What is a perceptron?

A. A basic computational unit that combines inputs using weights and an activation rule

B. A cloud storage service

C. An image file format

D. A network protocol

Correct Answer: A. A basic computational unit that combines inputs using weights and an activation rule

Why this is correct: A basic computational unit that combines inputs using weights and an activation rule directly addresses the concept tested in the question.

Why B is incorrect: A cloud storage service does not correctly answer the question.

Why C is incorrect: An image file format does not correctly answer the question.

Why D is incorrect: A network protocol does not correctly answer the question.

Question 2: What is a neural network weight?

A. A learned parameter that controls the contribution of an input

B. A dataset filename

C. A cloud region

D. A camera lens

Correct Answer: A. A learned parameter that controls the contribution of an input

Why this is correct: A learned parameter that controls the contribution of an input directly addresses the concept tested in the question.

Why B is incorrect: A dataset filename does not correctly answer the question.

Why C is incorrect: A cloud region does not correctly answer the question.

Why D is incorrect: A camera lens does not correctly answer the question.

Question 3: What is a bias term in a neural network?

A. A learnable offset added to a neuron's weighted input

B. A cloud password

C. A class label

D. A video frame

Correct Answer: A. A learnable offset added to a neuron's weighted input

Why this is correct: A learnable offset added to a neuron's weighted input directly addresses the concept tested in the question.

Why B is incorrect: A cloud password does not correctly answer the question.

Why C is incorrect: A class label does not correctly answer the question.

Why D is incorrect: A video frame does not correctly answer the question.

Question 4: Why are multiple layers useful in deep learning?

A. They allow models to learn increasingly complex representations

B. They eliminate the need for data

C. They guarantee zero error

D. They replace evaluation

Correct Answer: A. They allow models to learn increasingly complex representations

Why this is correct: They allow models to learn increasingly complex representations directly addresses the concept tested in the question.

Why B is incorrect: They eliminate the need for data does not correctly answer the question.

Why C is incorrect: They guarantee zero error does not correctly answer the question.

Why D is incorrect: They replace evaluation does not correctly answer the question.

Question 5: What is backpropagation used for?

A. Computing gradients used to update neural network parameters

B. Uploading files to OBS

C. Creating camera frames

D. Formatting a report

Correct Answer: A. Computing gradients used to update neural network parameters

Why this is correct: Computing gradients used to update neural network parameters directly addresses the concept tested in the question.

Why B is incorrect: Uploading files to OBS does not correctly answer the question.

Why C is incorrect: Creating camera frames does not correctly answer the question.

Why D is incorrect: Formatting a report does not correctly answer the question.

Question 6: What does an optimizer do during neural network training?

A. Updates model parameters using information from the loss and gradients

B. Stores image files permanently

C. Creates API endpoints automatically

D. Changes the model's class names

Correct Answer: A. Updates model parameters using information from the loss and gradients

Why this is correct: Updates model parameters using information from the loss and gradients directly addresses the concept tested in the question.

Why B is incorrect: Stores image files permanently does not correctly answer the question.

Why C is incorrect: Creates API endpoints automatically does not correctly answer the question.

Why D is incorrect: Changes the model's class names does not correctly answer the question.

Question 7: Why can ReLU be useful in hidden layers?

A. It provides a simple nonlinear activation and is computationally efficient

B. It is a cloud storage format

C. It performs OCR directly

D. It replaces the optimizer

Correct Answer: A. It provides a simple nonlinear activation and is computationally efficient

Why this is correct: It provides a simple nonlinear activation and is computationally efficient directly addresses the concept tested in the question.

Why B is incorrect: It is a cloud storage format does not correctly answer the question.

Why C is incorrect: It performs OCR directly does not correctly answer the question.

Why D is incorrect: It replaces the optimizer does not correctly answer the question.

Question 8: What is Sigmoid commonly useful for?

A. Producing an output bounded between 0 and 1 in suitable binary-output settings

B. Storing datasets

C. Detecting cameras

D. Managing API keys

Correct Answer: A. Producing an output bounded between 0 and 1 in suitable binary-output settings

Why this is correct: Producing an output bounded between 0 and 1 in suitable binary-output settings directly addresses the concept tested in the question.

Why B is incorrect: Storing datasets does not correctly answer the question.

Why C is incorrect: Detecting cameras does not correctly answer the question.

Why D is incorrect: Managing API keys does not correctly answer the question.

Question 9: What is Softmax commonly used for?

A. Converting class scores into a normalized distribution across multiple classes

B. Resizing images

C. Uploading models to OBS

D. Detecting network failures

Correct Answer: A. Converting class scores into a normalized distribution across multiple classes

Why this is correct: Converting class scores into a normalized distribution across multiple classes directly addresses the concept tested in the question.

Why B is incorrect: Resizing images does not correctly answer the question.

Why C is incorrect: Uploading models to OBS does not correctly answer the question.

Why D is incorrect: Detecting network failures does not correctly answer the question.

Question 10: Why is normalization often applied to neural-network inputs?

A. It can put input values on a more suitable numerical scale for training

B. It guarantees perfect accuracy

C. It removes labels

D. It replaces the loss function

Correct Answer: A. It can put input values on a more suitable numerical scale for training

Why this is correct: It can put input values on a more suitable numerical scale for training directly addresses the concept tested in the question.

Why B is incorrect: It guarantees perfect accuracy does not correctly answer the question.

Why C is incorrect: It removes labels does not correctly answer the question.

Why D is incorrect: It replaces the loss function does not correctly answer the question.

Question 11: What is a CNN feature hierarchy?

A. Early layers can learn simple patterns while deeper layers can learn more complex patterns

B. A list of cloud services

C. A database schema

D. A set of passwords

Correct Answer: A. Early layers can learn simple patterns while deeper layers can learn more complex patterns

Why this is correct: Early layers can learn simple patterns while deeper layers can learn more complex patterns directly addresses the concept tested in the question.

Why B is incorrect: A list of cloud services does not correctly answer the question.

Why C is incorrect: A database schema does not correctly answer the question.

Why D is incorrect: A set of passwords does not correctly answer the question.

Question 12: Why are pooling or downsampling operations used in CNNs?

A. They can reduce spatial computation and summarize local information

B. They create training labels

C. They increase image file size

D. They replace convolution

Correct Answer: A. They can reduce spatial computation and summarize local information

Why this is correct: They can reduce spatial computation and summarize local information directly addresses the concept tested in the question.

Why B is incorrect: They create training labels does not correctly answer the question.

Why C is incorrect: They increase image file size does not correctly answer the question.

Why D is incorrect: They replace convolution does not correctly answer the question.

Question 13: What does a convolution filter learn during training?

A. Weights that respond to useful local patterns in the input

B. Cloud API credentials

C. Dataset filenames

D. Operating-system settings

Correct Answer: A. Weights that respond to useful local patterns in the input

Why this is correct: Weights that respond to useful local patterns in the input directly addresses the concept tested in the question.

Why B is incorrect: Cloud API credentials does not correctly answer the question.

Why C is incorrect: Dataset filenames does not correctly answer the question.

Why D is incorrect: Operating-system settings does not correctly answer the question.

Question 14: Why is a validation curve useful during deep learning training?

A. It helps compare training behavior with performance on held-out data

B. It replaces the model

C. It changes GPU hardware

D. It creates OBS buckets

Correct Answer: A. It helps compare training behavior with performance on held-out data

Why this is correct: It helps compare training behavior with performance on held-out data directly addresses the concept tested in the question.

Why B is incorrect: It replaces the model does not correctly answer the question.

Why C is incorrect: It changes GPU hardware does not correctly answer the question.

Why D is incorrect: It creates OBS buckets does not correctly answer the question.

Question 15: What is a pretrained foundation model?

A. A model trained beforehand on broad data that can be reused or adapted for applications

B. A physical server

C. A database table

D. A camera sensor

Correct Answer: A. A model trained beforehand on broad data that can be reused or adapted for applications

Why this is correct: A model trained beforehand on broad data that can be reused or adapted for applications directly addresses the concept tested in the question.

Why B is incorrect: A physical server does not correctly answer the question.

Why C is incorrect: A database table does not correctly answer the question.

Why D is incorrect: A camera sensor does not correctly answer the question.

Question 16: What is fine-tuning?

A. Further training a pretrained model on task-specific data

B. Changing the monitor

C. Renaming the model file

D. Deleting the pretrained weights

Correct Answer: A. Further training a pretrained model on task-specific data

Why this is correct: Further training a pretrained model on task-specific data directly addresses the concept tested in the question.

Why B is incorrect: Changing the monitor does not correctly answer the question.

Why C is incorrect: Renaming the model file does not correctly answer the question.

Why D is incorrect: Deleting the pretrained weights does not correctly answer the question.

Question 17: Why might fine-tuning be useful?

A. It adapts existing learned representations to a particular task or dataset

B. It eliminates all evaluation

C. It guarantees no overfitting

D. It prevents deployment

Correct Answer: A. It adapts existing learned representations to a particular task or dataset

Why this is correct: It adapts existing learned representations to a particular task or dataset directly addresses the concept tested in the question.

Why B is incorrect: It eliminates all evaluation does not correctly answer the question.

Why C is incorrect: It guarantees no overfitting does not correctly answer the question.

Why D is incorrect: It prevents deployment does not correctly answer the question.

Question 18: Why compare TensorFlow and PyTorch?

A. Different frameworks have different APIs, abstractions, flexibility, and integration characteristics

B. They are both cloud storage services

C. They are image formats

D. They are database engines

Correct Answer: A. Different frameworks have different APIs, abstractions, flexibility, and integration characteristics

Why this is correct: Different frameworks have different APIs, abstractions, flexibility, and integration characteristics directly addresses the concept tested in the question.

Why B is incorrect: They are both cloud storage services does not correctly answer the question.

Why C is incorrect: They are image formats does not correctly answer the question.

Why D is incorrect: They are database engines does not correctly answer the question.

Question 19: What is a model checkpoint?

A. A saved state of a model, often used to resume training or preserve a selected version

B. A camera calibration file

C. A cloud bill

D. A CSV column

Correct Answer: A. A saved state of a model, often used to resume training or preserve a selected version

Why this is correct: A saved state of a model, often used to resume training or preserve a selected version directly addresses the concept tested in the question.

Why B is incorrect: A camera calibration file does not correctly answer the question.

Why C is incorrect: A cloud bill does not correctly answer the question.

Why D is incorrect: A CSV column does not correctly answer the question.

Question 20: Why inspect misclassified samples after training?

A. They can reveal patterns of errors and guide improvements to data or the model

B. They automatically retrain the model

C. They replace the test set

D. They increase GPU memory

Correct Answer: A. They can reveal patterns of errors and guide improvements to data or the model

Why this is correct: They can reveal patterns of errors and guide improvements to data or the model directly addresses the concept tested in the question.

Why B is incorrect: They automatically retrain the model does not correctly answer the question.

Why C is incorrect: They replace the test set does not correctly answer the question.

Why D is incorrect: They increase GPU memory does not correctly answer the question.

Phase 4 — Computer Vision (CV)

20 fresh questions in this phase.

Question 1: What information does a digital image contain at the pixel level?

A. Numerical intensity or color information

B. Only filenames

C. Only text labels

D. Only cloud metadata

Correct Answer: A. Numerical intensity or color information

Why this is correct: Numerical intensity or color information directly addresses the concept tested in the question.

Why B is incorrect: Only filenames does not correctly answer the question.

Why C is incorrect: Only text labels does not correctly answer the question.

Why D is incorrect: Only cloud metadata does not correctly answer the question.

Question 2: Why can color spaces be changed in computer vision?

A. Different color representations can make certain visual processing tasks easier

B. They always increase model accuracy

C. They replace object detection

D. They remove the need for images

Correct Answer: A. Different color representations can make certain visual processing tasks easier

Why this is correct: Different color representations can make certain visual processing tasks easier directly addresses the concept tested in the question.

Why B is incorrect: They always increase model accuracy does not correctly answer the question.

Why C is incorrect: They replace object detection does not correctly answer the question.

Why D is incorrect: They remove the need for images does not correctly answer the question.

Question 3: What is grayscale conversion?

A. Representing an image using intensity rather than full color channels

B. Converting an image into a cloud API

C. Creating a database

D. Adding more color channels

Correct Answer: A. Representing an image using intensity rather than full color channels

Why this is correct: Representing an image using intensity rather than full color channels directly addresses the concept tested in the question.

Why B is incorrect: Converting an image into a cloud API does not correctly answer the question.

Why C is incorrect: Creating a database does not correctly answer the question.

Why D is incorrect: Adding more color channels does not correctly answer the question.

Question 4: Why might image resizing be required before inference?

A. The model may expect a specific input spatial size

B. Resizing always improves accuracy

C. It creates labels automatically

D. It removes the need for preprocessing

Correct Answer: A. The model may expect a specific input spatial size

Why this is correct: The model may expect a specific input spatial size directly addresses the concept tested in the question.

Why B is incorrect: Resizing always improves accuracy does not correctly answer the question.

Why C is incorrect: It creates labels automatically does not correctly answer the question.

Why D is incorrect: It removes the need for preprocessing does not correctly answer the question.

Question 5: What is image normalization in a vision pipeline?

A. Scaling pixel values according to the model's expected numerical range or distribution

B. Deleting pixels randomly

C. Changing the object class

D. Creating a cloud bucket

Correct Answer: A. Scaling pixel values according to the model's expected numerical range or distribution

Why this is correct: Scaling pixel values according to the model's expected numerical range or distribution directly addresses the concept tested in the question.

Why B is incorrect: Deleting pixels randomly does not correctly answer the question.

Why C is incorrect: Changing the object class does not correctly answer the question.

Why D is incorrect: Creating a cloud bucket does not correctly answer the question.

Question 6: What is an image classification model's output?

A. A predicted class or class distribution for the image

B. Only a bounding box

C. Only camera coordinates

D. Only OCR cells

Correct Answer: A. A predicted class or class distribution for the image

Why this is correct: A predicted class or class distribution for the image directly addresses the concept tested in the question.

Why B is incorrect: Only a bounding box does not correctly answer the question.

Why C is incorrect: Only camera coordinates does not correctly answer the question.

Why D is incorrect: Only OCR cells does not correctly answer the question.

Question 7: What is the difference between image classification and object detection?

A. Classification labels an image, while detection also locates individual objects

B. Classification always needs video and detection never does

C. They are identical tasks

D. Detection cannot use neural networks

Correct Answer: A. Classification labels an image, while detection also locates individual objects

Why this is correct: Classification labels an image, while detection also locates individual objects directly addresses the concept tested in the question.

Why B is incorrect: Classification always needs video and detection never does does not correctly answer the question.

Why C is incorrect: They are identical tasks does not correctly answer the question.

Why D is incorrect: Detection cannot use neural networks does not correctly answer the question.

Question 8: What is IoU used for in object detection evaluation?

A. It measures overlap between predicted and reference regions

B. It measures network bandwidth

C. It measures CPU temperature

D. It measures OCR character count

Correct Answer: A. It measures overlap between predicted and reference regions

Why this is correct: It measures overlap between predicted and reference regions directly addresses the concept tested in the question.

Why B is incorrect: It measures network bandwidth does not correctly answer the question.

Why C is incorrect: It measures CPU temperature does not correctly answer the question.

Why D is incorrect: It measures OCR character count does not correctly answer the question.

Question 9: Why can multiple detections of the same object be a problem?

A. They can produce duplicate predictions for one object

B. They always improve precision

C. They reduce image resolution

D. They create more ground truth

Correct Answer: A. They can produce duplicate predictions for one object

Why this is correct: They can produce duplicate predictions for one object directly addresses the concept tested in the question.

Why B is incorrect: They always improve precision does not correctly answer the question.

Why C is incorrect: They reduce image resolution does not correctly answer the question.

Why D is incorrect: They create more ground truth does not correctly answer the question.

Question 10: What is Non-Maximum Suppression intended to address?

A. Selecting strong detections while suppressing highly overlapping duplicates

B. Removing all image noise

C. Training an RNN

D. Creating OCR tables

Correct Answer: A. Selecting strong detections while suppressing highly overlapping duplicates

Why this is correct: Selecting strong detections while suppressing highly overlapping duplicates directly addresses the concept tested in the question.

Why B is incorrect: Removing all image noise does not correctly answer the question.

Why C is incorrect: Training an RNN does not correctly answer the question.

Why D is incorrect: Creating OCR tables does not correctly answer the question.

Question 11: What is semantic segmentation?

A. Assigning a semantic class to image pixels

B. Drawing only one box per image

C. Classifying the entire dataset

D. Extracting audio

Correct Answer: A. Assigning a semantic class to image pixels

Why this is correct: Assigning a semantic class to image pixels directly addresses the concept tested in the question.

Why B is incorrect: Drawing only one box per image does not correctly answer the question.

Why C is incorrect: Classifying the entire dataset does not correctly answer the question.

Why D is incorrect: Extracting audio does not correctly answer the question.

Question 12: What is instance segmentation?

A. Separating individual object instances with their own masks

B. Assigning one label to every image

C. Detecting only edges

D. Changing RGB to grayscale

Correct Answer: A. Separating individual object instances with their own masks

Why this is correct: Separating individual object instances with their own masks directly addresses the concept tested in the question.

Why B is incorrect: Assigning one label to every image does not correctly answer the question.

Why C is incorrect: Detecting only edges does not correctly answer the question.

Why D is incorrect: Changing RGB to grayscale does not correctly answer the question.

Question 13: Why are masks useful in segmentation?

A. They identify which pixels belong to a region, class, or object

B. They store cloud credentials

C. They replace datasets

D. They measure CPU speed

Correct Answer: A. They identify which pixels belong to a region, class, or object

Why this is correct: They identify which pixels belong to a region, class, or object directly addresses the concept tested in the question.

Why B is incorrect: They store cloud credentials does not correctly answer the question.

Why C is incorrect: They replace datasets does not correctly answer the question.

Why D is incorrect: They measure CPU speed does not correctly answer the question.

Question 14: Why can OCR preprocessing improve document recognition?

A. Cleaner and appropriately prepared input can make text extraction easier

B. Preprocessing always guarantees perfect OCR

C. It eliminates the OCR model

D. It changes the document's legal meaning

Correct Answer: A. Cleaner and appropriately prepared input can make text extraction easier

Why this is correct: Cleaner and appropriately prepared input can make text extraction easier directly addresses the concept tested in the question.

Why B is incorrect: Preprocessing always guarantees perfect OCR does not correctly answer the question.

Why C is incorrect: It eliminates the OCR model does not correctly answer the question.

Why D is incorrect: It changes the document's legal meaning does not correctly answer the question.

Question 15: What is table OCR different from ordinary plain-text OCR?

A. It aims to preserve or extract structured table information such as cells and values

B. It only detects faces

C. It only detects edges

D. It only converts video

Correct Answer: A. It aims to preserve or extract structured table information such as cells and values

Why this is correct: It aims to preserve or extract structured table information such as cells and values directly addresses the concept tested in the question.

Why B is incorrect: It only detects faces does not correctly answer the question.

Why C is incorrect: It only detects edges does not correctly answer the question.

Why D is incorrect: It only converts video does not correctly answer the question.

Question 16: Why should an OCR API be tested with varied documents?

A. Different layouts and image conditions can reveal strengths and failure cases

B. Testing one image is always sufficient

C. Variation prevents evaluation

D. It changes the API credentials

Correct Answer: A. Different layouts and image conditions can reveal strengths and failure cases

Why this is correct: Different layouts and image conditions can reveal strengths and failure cases directly addresses the concept tested in the question.

Why B is incorrect: Testing one image is always sufficient does not correctly answer the question.

Why C is incorrect: Variation prevents evaluation does not correctly answer the question.

Why D is incorrect: It changes the API credentials does not correctly answer the question.

Question 17: What is a camera frame in video analytics?

A. A single image captured from a video stream at a particular time

B. A database record only

C. A cloud password

D. A neural-network weight

Correct Answer: A. A single image captured from a video stream at a particular time

Why this is correct: A single image captured from a video stream at a particular time directly addresses the concept tested in the question.

Why B is incorrect: A database record only does not correctly answer the question.

Why C is incorrect: A cloud password does not correctly answer the question.

Why D is incorrect: A neural-network weight does not correctly answer the question.

Question 18: What is object tracking?

A. Following an identified object's location across video frames

B. Classifying one static image only

C. Converting speech to text

D. Storing a model in OBS

Correct Answer: A. Following an identified object's location across video frames

Why this is correct: Following an identified object's location across video frames directly addresses the concept tested in the question.

Why B is incorrect: Classifying one static image only does not correctly answer the question.

Why C is incorrect: Converting speech to text does not correctly answer the question.

Why D is incorrect: Storing a model in OBS does not correctly answer the question.

Question 19: Why can edge inference reduce response delay?

A. Processing can occur closer to the source of visual data

B. Edge devices always have more compute than clouds

C. It removes all model computation

D. It guarantees zero network traffic

Correct Answer: A. Processing can occur closer to the source of visual data

Why this is correct: Processing can occur closer to the source of visual data directly addresses the concept tested in the question.

Why B is incorrect: Edge devices always have more compute than clouds does not correctly answer the question.

Why C is incorrect: It removes all model computation does not correctly answer the question.

Why D is incorrect: It guarantees zero network traffic does not correctly answer the question.

Question 20: Why is confidence threshold selection important in detection systems?

A. It controls which predicted detections are accepted based on model confidence

B. It changes the camera lens

C. It creates training labels

D. It increases image dimensions

Correct Answer: A. It controls which predicted detections are accepted based on model confidence

Why this is correct: It controls which predicted detections are accepted based on model confidence directly addresses the concept tested in the question.

Why B is incorrect: It changes the camera lens does not correctly answer the question.

Why C is incorrect: It creates training labels does not correctly answer the question.

Why D is incorrect: It increases image dimensions does not correctly answer the question.

Phase 5 — Huawei Cloud + DevOps/MLOps

20 fresh questions in this phase.

Question 1: What is the main role of ModelArts in the AI lifecycle?

A. Supporting activities such as dataset management, training, evaluation, and deployment

B. Replacing all programming languages

C. Managing office printers

D. Editing PDFs

Correct Answer: A. Supporting activities such as dataset management, training, evaluation, and deployment

Why this is correct: Supporting activities such as dataset management, training, evaluation, and deployment directly addresses the concept tested in the question.

Why B is incorrect: Replacing all programming languages does not correctly answer the question.

Why C is incorrect: Managing office printers does not correctly answer the question.

Why D is incorrect: Editing PDFs does not correctly answer the question.

Question 2: What is an AI notebook environment useful for?

A. Experimenting with code, data, and models interactively

B. Replacing model deployment

C. Managing keyboard drivers

D. Changing network cables

Correct Answer: A. Experimenting with code, data, and models interactively

Why this is correct: Experimenting with code, data, and models interactively directly addresses the concept tested in the question.

Why B is incorrect: Replacing model deployment does not correctly answer the question.

Why C is incorrect: Managing keyboard drivers does not correctly answer the question.

Why D is incorrect: Changing network cables does not correctly answer the question.

Question 3: Why is cloud object storage useful for AI datasets?

A. It provides a centralized location for storing and accessing data used by AI workflows

B. It performs model inference automatically

C. It replaces all APIs

D. It trains every model automatically

Correct Answer: A. It provides a centralized location for storing and accessing data used by AI workflows

Why this is correct: It provides a centralized location for storing and accessing data used by AI workflows directly addresses the concept tested in the question.

Why B is incorrect: It performs model inference automatically does not correctly answer the question.

Why C is incorrect: It replaces all APIs does not correctly answer the question.

Why D is incorrect: It trains every model automatically does not correctly answer the question.

Question 4: What does a model deployment endpoint provide?

A. A network-accessible interface through which an application can request predictions

B. A physical camera

C. A training dataset

D. A Python interpreter

Correct Answer: A. A network-accessible interface through which an application can request predictions

Why this is correct: A network-accessible interface through which an application can request predictions directly addresses the concept tested in the question.

Why B is incorrect: A physical camera does not correctly answer the question.

Why C is incorrect: A training dataset does not correctly answer the question.

Why D is incorrect: A Python interpreter does not correctly answer the question.

Question 5: What should be protected when integrating a cloud AI API?

A. Authentication credentials and access keys

B. Only image filenames

C. Only monitor settings

D. Only notebook titles

Correct Answer: A. Authentication credentials and access keys

Why this is correct: Authentication credentials and access keys directly addresses the concept tested in the question.

Why B is incorrect: Only image filenames does not correctly answer the question.

Why C is incorrect: Only monitor settings does not correctly answer the question.

Why D is incorrect: Only notebook titles does not correctly answer the question.

Question 6: Why should API credentials be configured securely?

A. Exposed credentials can allow unauthorized access to cloud resources or services

B. Security only affects image colors

C. Credentials are not sensitive

D. Secure configuration prevents API use

Correct Answer: A. Exposed credentials can allow unauthorized access to cloud resources or services

Why this is correct: Exposed credentials can allow unauthorized access to cloud resources or services directly addresses the concept tested in the question.

Why B is incorrect: Security only affects image colors does not correctly answer the question.

Why C is incorrect: Credentials are not sensitive does not correctly answer the question.

Why D is incorrect: Secure configuration prevents API use does not correctly answer the question.

Question 7: What is an HTTP status code useful for during API integration?

A. It indicates the result or status of an HTTP request

B. It stores model weights

C. It labels images

D. It measures GPU temperature

Correct Answer: A. It indicates the result or status of an HTTP request

Why this is correct: It indicates the result or status of an HTTP request directly addresses the concept tested in the question.

Why B is incorrect: It stores model weights does not correctly answer the question.

Why C is incorrect: It labels images does not correctly answer the question.

Why D is incorrect: It measures GPU temperature does not correctly answer the question.

Question 8: Why should API clients implement error handling?

A. Network, authentication, input, timeout, and service errors can occur

B. Successful requests are the only possible outcome

C. Error handling changes model architecture

D. It eliminates the need for logs

Correct Answer: A. Network, authentication, input, timeout, and service errors can occur

Why this is correct: Network, authentication, input, timeout, and service errors can occur directly addresses the concept tested in the question.

Why B is incorrect: Successful requests are the only possible outcome does not correctly answer the question.

Why C is incorrect: Error handling changes model architecture does not correctly answer the question.

Why D is incorrect: It eliminates the need for logs does not correctly answer the question.

Question 9: What is CI/CD intended to automate?

A. Building, testing, and delivering software changes through a repeatable pipeline

B. Only image labeling

C. Only OCR extraction

D. Only hardware installation

Correct Answer: A. Building, testing, and delivering software changes through a repeatable pipeline

Why this is correct: Building, testing, and delivering software changes through a repeatable pipeline directly addresses the concept tested in the question.

Why B is incorrect: Only image labeling does not correctly answer the question.

Why C is incorrect: Only OCR extraction does not correctly answer the question.

Why D is incorrect: Only hardware installation does not correctly answer the question.

Question 10: How does MLOps extend traditional DevOps ideas?

A. It applies lifecycle practices to data, models, experiments, and ML deployments

B. It removes version control

C. It only manages hardware

D. It prevents automated testing

Correct Answer: A. It applies lifecycle practices to data, models, experiments, and ML deployments

Why this is correct: It applies lifecycle practices to data, models, experiments, and ML deployments directly addresses the concept tested in the question.

Why B is incorrect: It removes version control does not correctly answer the question.

Why C is incorrect: It only manages hardware does not correctly answer the question.

Why D is incorrect: It prevents automated testing does not correctly answer the question.

Question 11: Why is automated testing useful in an ML pipeline?

A. It can detect regressions in data processing, APIs, and system behavior

B. It guarantees model accuracy

C. It removes the need for validation

D. It replaces monitoring

Correct Answer: A. It can detect regressions in data processing, APIs, and system behavior

Why this is correct: It can detect regressions in data processing, APIs, and system behavior directly addresses the concept tested in the question.

Why B is incorrect: It guarantees model accuracy does not correctly answer the question.

Why C is incorrect: It removes the need for validation does not correctly answer the question.

Why D is incorrect: It replaces monitoring does not correctly answer the question.

Question 12: What is integration testing?

A. Testing whether multiple components work correctly together

B. Testing only one Python function

C. Testing only monitor hardware

D. Testing only the final PDF

Correct Answer: A. Testing whether multiple components work correctly together

Why this is correct: Testing whether multiple components work correctly together directly addresses the concept tested in the question.

Why B is incorrect: Testing only one Python function does not correctly answer the question.

Why C is incorrect: Testing only monitor hardware does not correctly answer the question.

Why D is incorrect: Testing only the final PDF does not correctly answer the question.

Question 13: What is system testing?

A. Testing the complete application or pipeline against expected behavior

B. Testing only a single variable

C. Testing only a model layer

D. Testing only a cloud password

Correct Answer: A. Testing the complete application or pipeline against expected behavior

Why this is correct: Testing the complete application or pipeline against expected behavior directly addresses the concept tested in the question.

Why B is incorrect: Testing only a single variable does not correctly answer the question.

Why C is incorrect: Testing only a model layer does not correctly answer the question.

Why D is incorrect: Testing only a cloud password does not correctly answer the question.

Question 14: What is a pipeline trigger?

A. An event or condition that starts an automated workflow

B. A database column

C. A monitor setting

D. A model label

Correct Answer: A. An event or condition that starts an automated workflow

Why this is correct: An event or condition that starts an automated workflow directly addresses the concept tested in the question.

Why B is incorrect: A database column does not correctly answer the question.

Why C is incorrect: A monitor setting does not correctly answer the question.

Why D is incorrect: A model label does not correctly answer the question.

Question 15: Why can event-driven automation be useful in cloud AI?

A. A data or inference event can automatically start a defined follow-up action

B. It prevents automation

C. It requires every action to be manual

D. It removes the need for APIs

Correct Answer: A. A data or inference event can automatically start a defined follow-up action

Why this is correct: A data or inference event can automatically start a defined follow-up action directly addresses the concept tested in the question.

Why B is incorrect: It prevents automation does not correctly answer the question.

Why C is incorrect: It requires every action to be manual does not correctly answer the question.

Why D is incorrect: It removes the need for APIs does not correctly answer the question.

Question 16: What is monitoring in an AI production system?

A. Observing system and model behavior after deployment

B. Training only once

C. Deleting logs

D. Changing image formats

Correct Answer: A. Observing system and model behavior after deployment

Why this is correct: Observing system and model behavior after deployment directly addresses the concept tested in the question.

Why B is incorrect: Training only once does not correctly answer the question.

Why C is incorrect: Deleting logs does not correctly answer the question.

Why D is incorrect: Changing image formats does not correctly answer the question.

Question 17: Why monitor model latency?

A. High latency can violate application response-time requirements

B. Latency changes model labels

C. Latency creates new training data automatically

D. Latency measures image color

Correct Answer: A. High latency can violate application response-time requirements

Why this is correct: High latency can violate application response-time requirements directly addresses the concept tested in the question.

Why B is incorrect: Latency changes model labels does not correctly answer the question.

Why C is incorrect: Latency creates new training data automatically does not correctly answer the question.

Why D is incorrect: Latency measures image color does not correctly answer the question.

Question 18: Why monitor input-data distributions?

A. Changes can indicate drift that may affect model performance

B. They change monitor brightness

C. They replace ground truth

D. They eliminate retraining

Correct Answer: A. Changes can indicate drift that may affect model performance

Why this is correct: Changes can indicate drift that may affect model performance directly addresses the concept tested in the question.

Why B is incorrect: They change monitor brightness does not correctly answer the question.

Why C is incorrect: They replace ground truth does not correctly answer the question.

Why D is incorrect: They eliminate retraining does not correctly answer the question.

Question 19: What is a data-drift-triggered retraining workflow?

A. Detect drift, trigger training, evaluate the updated model, and deploy when appropriate

B. Delete the old dataset immediately

C. Disable monitoring

D. Deploy every model without evaluation

Correct Answer: A. Detect drift, trigger training, evaluate the updated model, and deploy when appropriate

Why this is correct: Detect drift, trigger training, evaluate the updated model, and deploy when appropriate directly addresses the concept tested in the question.

Why B is incorrect: Delete the old dataset immediately does not correctly answer the question.

Why C is incorrect: Disable monitoring does not correctly answer the question.

Why D is incorrect: Deploy every model without evaluation does not correctly answer the question.

Question 20: Why are logs important in MLOps?

A. They provide evidence for troubleshooting, monitoring, auditing, and performance analysis

B. They replace source code

C. They guarantee zero errors

D. They remove the need for testing

Correct Answer: A. They provide evidence for troubleshooting, monitoring, auditing, and performance analysis

Why this is correct: They provide evidence for troubleshooting, monitoring, auditing, and performance analysis directly addresses the concept tested in the question.

Why B is incorrect: They replace source code does not correctly answer the question.

Why C is incorrect: They guarantee zero errors does not correctly answer the question.

Why D is incorrect: They remove the need for testing does not correctly answer the question.

Final Answer Key — 2nd Edition

Phase 1 — Artificial Intelligence (AI)

1. A 2. A 3. A 4. A 5. A 6. A 7. A 8. A 9. A 10. A

11. A 12. A 13. A 14. A 15. A 16. A 17. A 18. A 19. A 20. A

Phase 2 — Machine Learning (ML)

21. A 22. A 23. A 24. A 25. A 26. A 27. A 28. A 29. A 30. A

31. A 32. A 33. A 34. A 35. A 36. A 37. A 38. A 39. A 40. A

Phase 3 — Deep Learning (DL)

41. A 42. A 43. A 44. A 45. A 46. A 47. A 48. A 49. A 50. A

51. A 52. A 53. A 54. A 55. A 56. A 57. A 58. A 59. A 60. A

Phase 4 — Computer Vision (CV)

61. A 62. A 63. A 64. A 65. A 66. A 67. A 68. A 69. A 70. A

71. A 72. A 73. A 74. A 75. A 76. A 77. A 78. A 79. A 80. A

Phase 5 — Huawei Cloud + DevOps/MLOps

81. A 82. A 83. A 84. A 85. A 86. A 87. A 88. A 89. A 90. A

91. A 92. A 93. A 94. A 95. A 96. A 97. A 98. A 99. A 100. A